Histogram Equalization

Assuming initially continuous intensity values, let the variable *r* denote the intensities of an image to be processed. As usual, we assume that *r* is in the range $[0, L-1]$, with $r = 0$ representing black and $r = L-1$ representing white. For *r* satisfying these conditions, we focus attention on transformations (intensity mappings) of the form

$$s = T(r) \quad 0 \le r \le L-1 \tag{3-8}$$

that produce an output intensity value, *s*, for a given intensity value *r* in the input image. We assume that

a. *T*(*r*) is a monotonic† increasing function in the interval $0 \le r \le L-1$; and

† A function *T*(*r*) is a *monotonic increasing* function if $T(r_2) \ge T(r_1)$ for $r_2 > r_1$. $T(r)$ is a *strictly monotonic increasing* function if $T(r_2) > T(r_1)$ for $r_2 > r_1$. Similar definitions apply to a monotonic decreasing function.

b. $0 \le T(r) \le L-1$ for $0 \le r \le L-1$.

a b

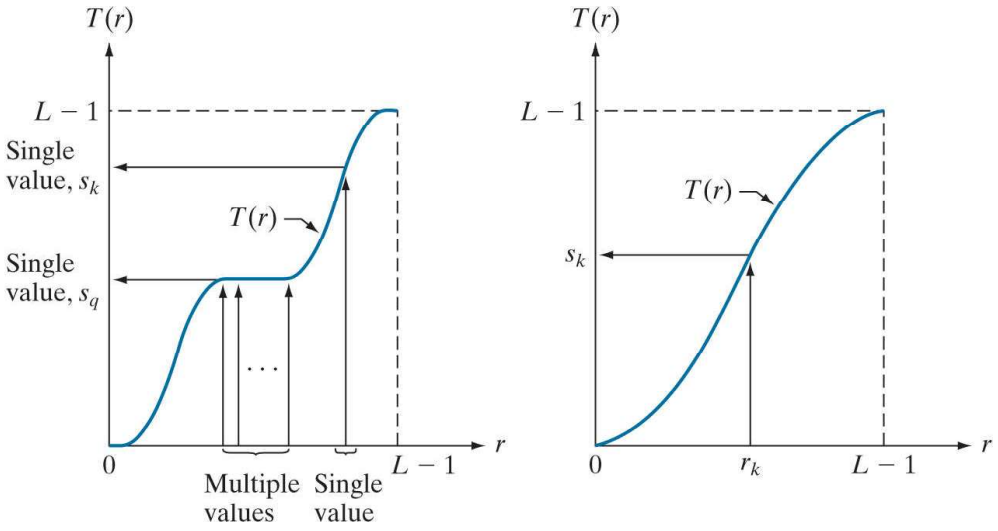


FIGURE 3.17
(a) Monotonic increasing function, showing how multiple values can map to a single value. (b) Strictly monotonic increasing function. This is a one-to-one mapping, both ways.

In some formulations to be discussed shortly, we use the inverse transformation

$$r = T^{-1}(s) \quad 0 \le s \le L-1 \tag{3-9}$$

in which case we change condition (a) to:

(a′) *T*(*r*) is a *strictly* monotonic increasing function in the interval $0 \le r \le L-1$.

The condition in (a) that *T*(*r*). be monotonically increasing guarantees that output intensity values will never be less than corresponding input values, thus preventing artifacts created by reversals of intensity. Condition (b) guarantees that the range of output intensities is the same as the input. Finally, condition (a') guarantees that the mappings from *s* back to *r* will be one-to-one, thus preventing ambiguities.

Figure 3.17(a) shows a function that satisfies conditions (a) and (b). Here, we see that it is possible for multiple input values to map to a single output value and still satisfy these two conditions. That is, a monotonic transformation function performs a one-to-one or many-to-one mapping. This is perfectly fine when mapping from *r* to *s*. However, Fig. 3.17(a) presents a problem if we wanted to recover the values of *r* uniquely from the mapped values (inverse mapping can be visualized by reversing the direction of the arrows). This would be possible for the inverse mapping of s_k in Fig. 3.17(a), but the inverse mapping of s_q is a range of values, which, of

course, prevents us in general from recovering the original value of r that resulted in s_q. As Fig. 3.17(b) shows, requiring that $T(r)$ be strictly monotonic guarantees that the inverse mappings will be *single valued* (i.e., the mapping is one-to-one in both directions).This is a theoretical requirement that will allow us to derive some important histogram processing techniques later in this chapter. Because images are stored using integer intensity values, we are forced to round all results to their nearest integer values. This often results in strict monotonicity not being satisfied, which implies inverse transformations that may not be unique. Fortunately, this problem is not difficult to handle in the discrete case, as Example 3.7 in this section illustrates.

As mentioned in Section 2.6, the intensity of an image may be viewed as a random variable in the interval $[0, L-1]$. Let $p_r(r)$ and $p_s(s)$ denote the PDFs of intensity values r and s in two different images. The subscripts on p indicate that p_r and p_s are different functions. A fundamental result from probability theory is that if $p_r(r)$ and $T(r)$ are known, and $T(r)$ is continuous and differentiable over the range of values of interest, then the PDF of the transformed (mapped) variable s can be obtained as

$$p_s(s) = p_r(r)\left|\frac{dr}{ds}\right| \tag{3-10}$$

Thus, we see that the PDF of the output intensity variable, s, is determined by the PDF of the input intensities and the transformation function used [recall that r and s are related by $T(r)$].

A transformation function of particular importance in image processing is

$$s = T(r) = (L-1)\int_0^r p_r(w)dw \tag{3-11}$$

where w is a dummy variable of integration. The integral on the right side is the *cumulative distribution function* (CDF) of random variable r (see Section 2.6). Because PDFs always are positive, and the integral of a function is the area under the function, it follows that the transformation function of Eq. (3-11) satisfies condition (a). This is because the area under the function cannot decrease as r increases. When the upper limit in this equation is $r = (L-1)$ the integral evaluates to 1, as it must for a PDF. Thus, the maximum value of s is $L-1$, and condition (b) is satisfied also.

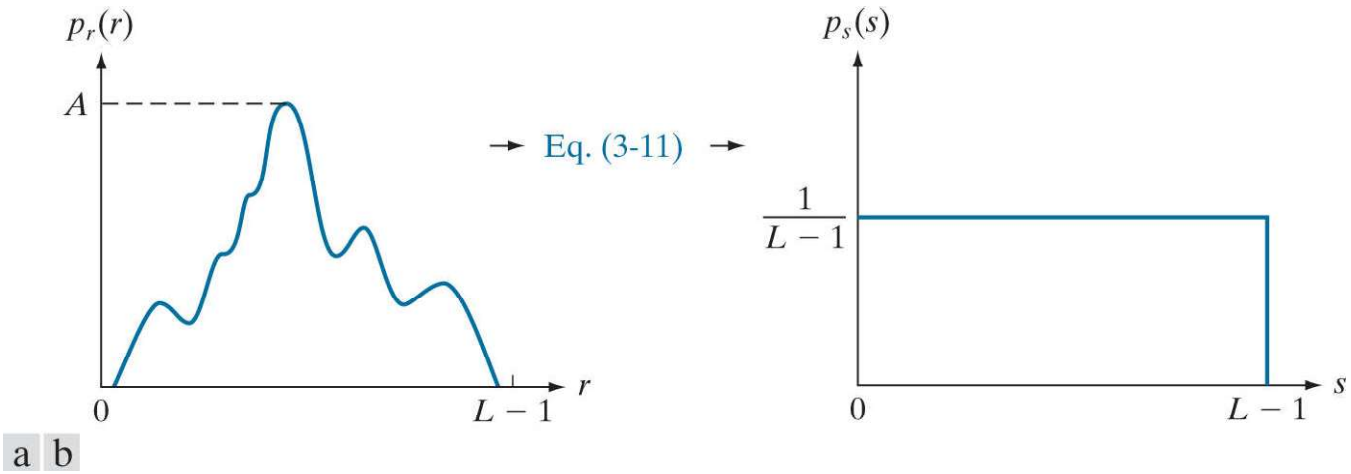
We use Eq. (3-10) to find the $p_s(s)$ corresponding to the transformation just discussed. We know from Leibniz's rule in calculus that the derivative of a definite integral with respect to its upper limit is the integrand evaluated at the limit. That is,

$$\begin{aligned}\frac{ds}{dr} &= \frac{dT(r)}{dr} \\ &= (L-1)\frac{d}{dr}\left[\int_0^r p_r(w)dw\right] \\ &= (L-1)p_r(r)\end{aligned} \tag{3-12}$$

Substituting this result for dr/ds in Eq. (3-10), and noting that all probability values are positive, gives the result

$$\begin{aligned}p_s(s) &= p_r(r)\left|\frac{dr}{ds}\right| \\ &= p_r(r)\left|\frac{1}{(L-1)p_r(r)}\right| \\ &= \frac{1}{L-1} \quad 0 \le s \le L-1\end{aligned} \tag{3-13}$$

We recognize the form of $p_s(s)$ in the last line of this equation as a *uniform* probability density function. Thus, performing the intensity transformation in Eq. (3-11) yields a random variable, s, characterized by a uniform PDF. What is important is that $p_s(s)$ in Eq. (3-13) will *always* be uniform, *independently* of the form of $p_r(r)$. Figure 3.18 and the following example illustrate these concepts.



a b

FIGURE 3.18
(a) An arbitrary PDF. (b) Result of applying Eq. (3-11) to the input PDF. The resulting PDF is always uniform, independently of the shape of the input.

Example 3.4: Illustration of Eqs. (3-11) and (3-13) .

Suppose that the (continuous) intensity values in an image have the PDF

$$P_r(r) = \begin{cases} \frac{2r}{(L-1)^2} & \text{for } 0 \le r \le L-1 \\ 0 & \text{otherwise} \end{cases}$$

From Eq. (3-11)

$$s = T(r) = (L-1)\int_0^r p_r(w)dw = \frac{2}{L-1}\int_0^r w dw = \frac{r^2}{L-1}$$

Suppose that we form a new image with intensities, s, obtained using this transformation; that is, the s values are formed by squaring the corresponding intensity values of the input image, then dividing them by $L-1$. We can verify that the PDF of the intensities in the new image, $p_s(s)$, is uniform by substituting $p_r(r)$ into Eq. (3-13) , and using the fact that $s = r^2/(L-1)$; that is,

$$p_s(s) = p_r(r)\left|\frac{dr}{ds}\right| = \frac{2r}{(L-1)^2}\left|\left[\frac{ds}{dr}\right]^{-1}\right|$$
$$= \frac{2r}{(L-1)^2}\left|\left[\frac{d}{dr}\frac{r^2}{L-1}\right]^{-1}\right| = \frac{2r}{(L-1)^2}\left|\frac{(L-1)}{2r}\right| = \frac{1}{L-1}$$

The last step follows because r is nonnegative and $L > 1$. As expected, the result is a uniform PDF.

For discrete values, we work with probabilities and summations instead of probability density functions and integrals (but the requirement of monotonicity stated earlier still applies). Recall that the probability of occurrence of intensity level r_k in a digital image is approximated by

$$P_r(r_k) = \frac{n_k}{MN} \tag{3-14}$$

where MN is the total number of pixels in the image, and n_k denotes the number of pixels that have intensity r_k. As noted in the beginning of this section, $p_r(r_k)$, with $r_k \in [0, L-1]$, is commonly referred to as a normalized image histogram.

The discrete form of the transformation in Eq. (3-11) is

$$s_k = T(r_k) = (L-1)\sum_{j=0}^{k} p_r(r_j) \quad k = 0,\ 1,\ 2,\ \ldots,\ L-1 \tag{3-15}$$

where, as before, *L* is the number of possible intensity levels in the image (e.g., 256 for an 8-bit image). Thus, a processed (output) image is obtained by using **Eq. (3-15)** to map each pixel in the input image with intensity r_k into a corresponding pixel with level s_k in the output image, This is called a *histogram equalization* or *histogram linearization* transformation. It is not difficult to show (see **Problem 3.11**) that this transformation satisfies conditions (a) and (b) stated previously in this section.